

5. MMANA-GAL BASED ANTENNA DESIGN

5.1 Design and Optimization of Wire Antenna using MMANA-GAL

INPUT VALUES:

MMANA-GALbasic C:\MMANA-GALBasic3\yagiuda_antenna_5.maa

File Edit Tools Setup Help MMANA-GALpro

Geometry View Calculate Far field plots

Name

Freq MHz ☐ lambda

Wires 5 Auto segmentation: DM1 DM2 SC EC ☐ Keep connect.

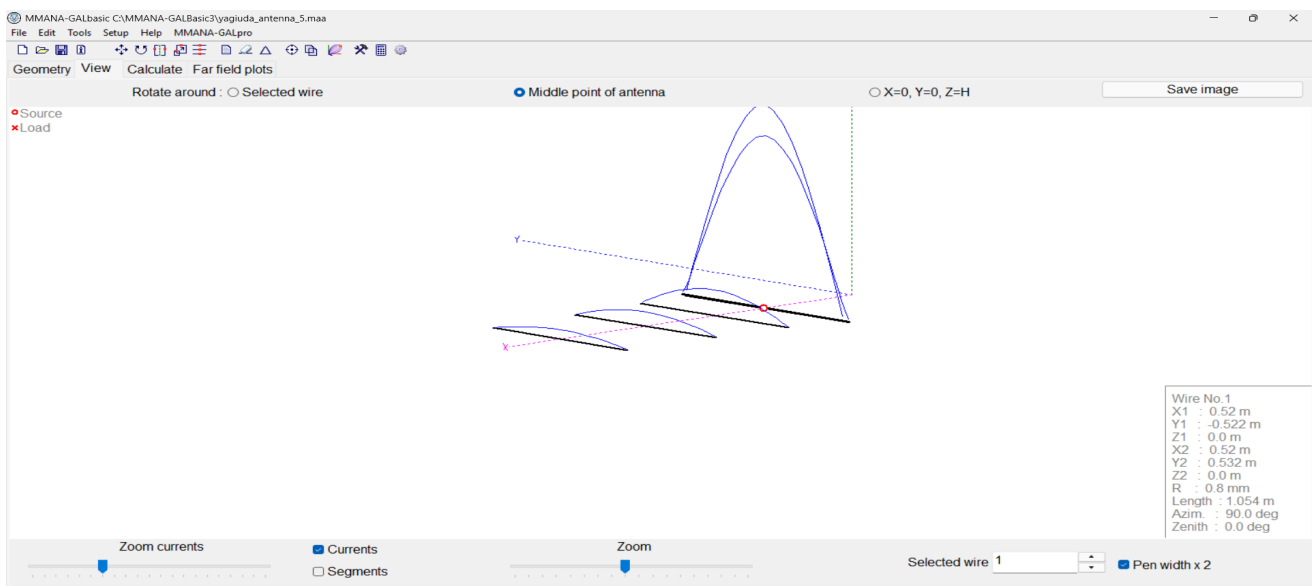
No.	X1(m)	Y1(m)	Z1(m)	X2(m)	Y2(m)	Z2(m)	R(mm)	Seg.
1	0.52	-0.522	0.0	0.52	0.532	0.0	0.8	-1
2	0.532	-0.497	0.0	0.532	0.497	0.0	0.8	-1
3	0.838	-0.472	0.0	0.838	0.472	0.0	0.8	-1
4	1.257	-0.449	0.0	1.257	0.449	0.0	0.8	-1
5	1.78	-0.426	0.0	1.78	0.426	0.0	0.8	-1
next								

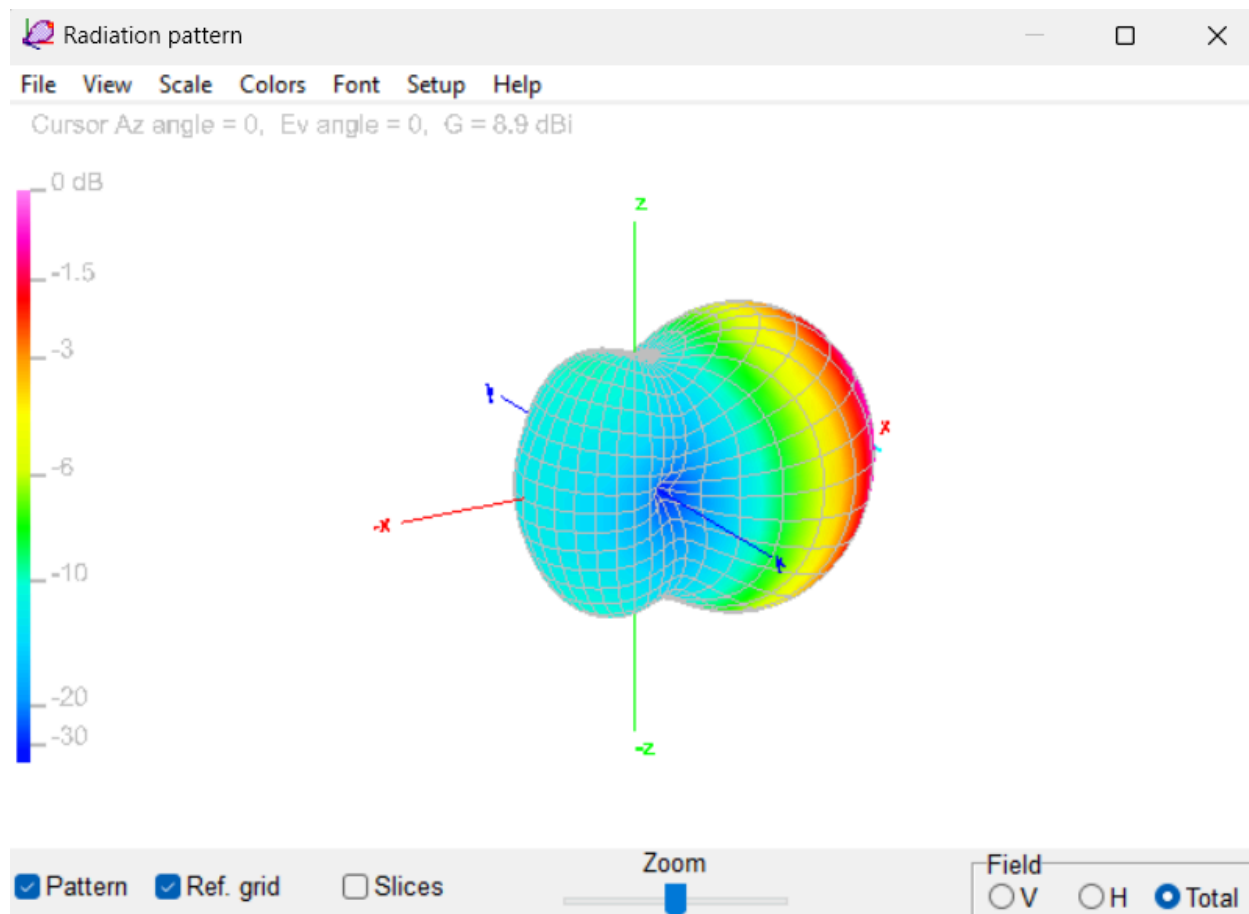
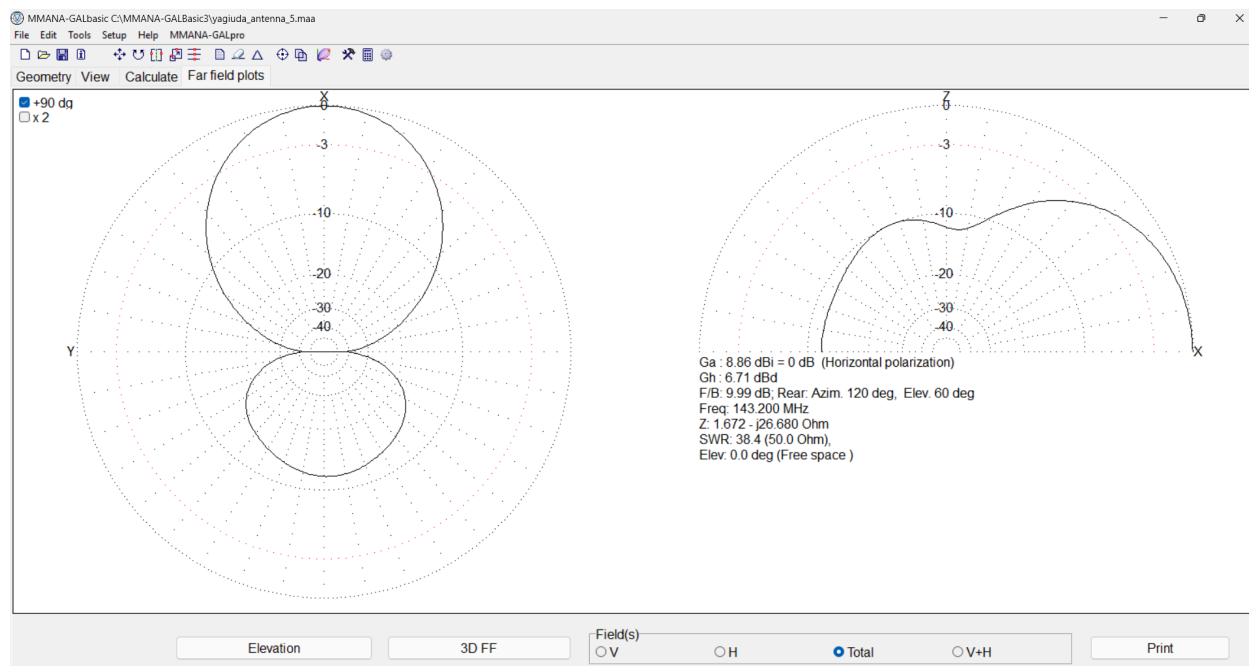
Sources 1

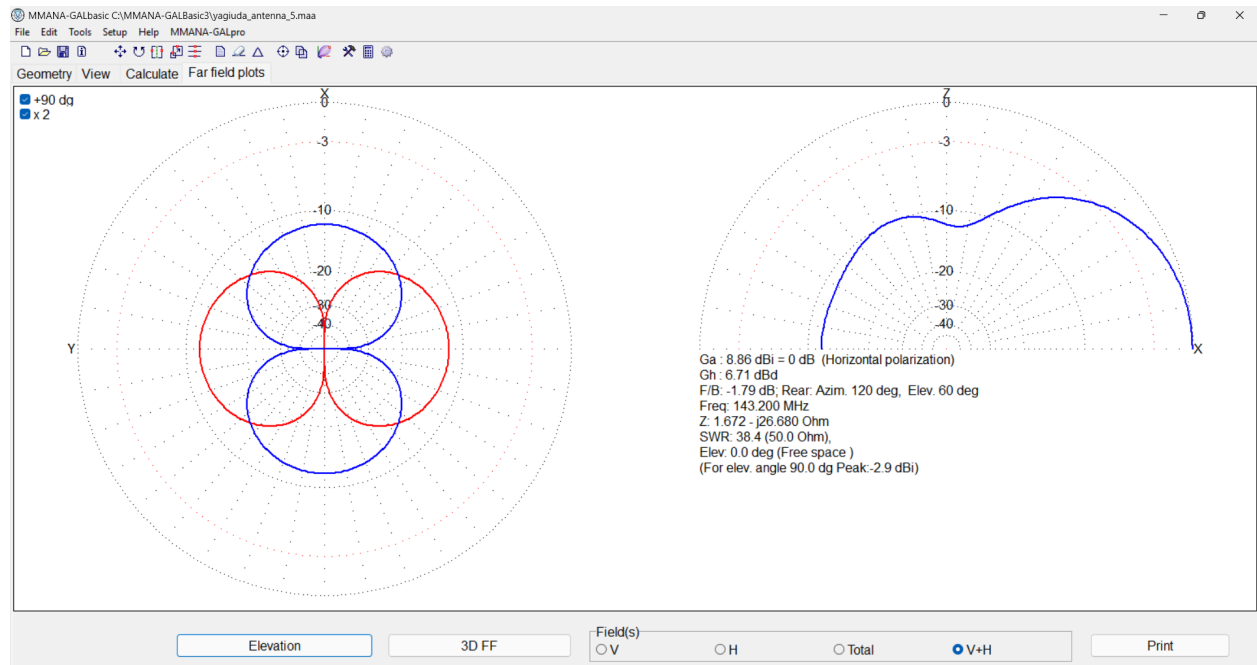
No.	PULSE	Volt. V	Phase dg
1	w2c	10.0	0.0
next			

Loads 0 (L - uH; C - pF; R/jX - Ohm) ☐ Use loads

No.	PULSE	Type	L/R/A0	C/jX/B0	Q/A1	F/B1
next						







5.2 Design and Optimization of Yagi-Uda Antenna using MMANA-GAL

MMANA-GALbasic C:\MMANA-GALbasic3\wira_antenna.maa

File Edit Tools Setup Help MMANA-GALpro

Geometry View Calculate Far field plots

Name

Freq MHz ☐ lambda

Wires 1 Auto segmentation: DM1 DM2 SC EC ☐ Keep connect.

No.	X1(m)	Y1(m)	Z1(m)	X2(m)	Y2(m)	Z2(m)	R(mm)	Seg.
1	0.0	-0.524	0.0	0.0	0.524	0.0	0.8	-1
next								

Sources 1

No.	PULSE	Volt. V	Phase dg
1	w1c	10.0	0.0
next			

Loads 0 (L - uH; C - pF; R/jX - Ohm) ☐ Use loads

No.	PULSE	Type	L/R/A0	C/jX/B0	Q/A1	F/B1
next						

MMANA-GALbasic C:\MMANA-GALbasic3\wira_antenna.maa

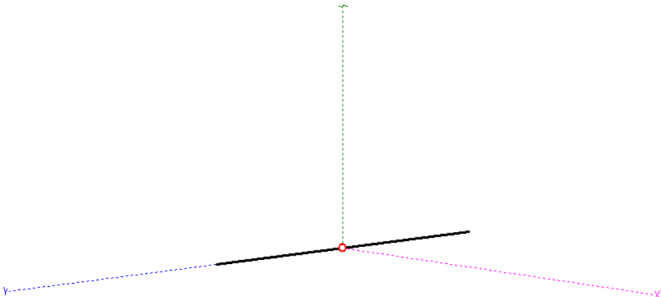
File Edit Tools Setup Help MMANA-GALpro

Geometry View Calculate Far field plots

Rotate around : ☐ Selected wire ☒ Middle point of antenna ☐ X=0, Y=0, Z=H

Source

Load



Zoom currents

☒ Currents ☐ Segments

Zoom

Selected wire ☒ Pen width x 2

Wire No. 1

X1 : 0.0 m

Y1 : -0.524 m

Z1 : 0.0 m

X2 : 0.0 m

Y2 : 0.524 m

Z2 : 0.0 m

R : 0.8 mm

Length : 1.048 m

Azim. : 90.0 deg

Zenith : 0.0 deg

